The world brane of 7+1 spacetime: symbols and their definitions

A dictionary of every symbol used in the document on the world brane, with its defining equation and its name in the notebook 7+1.nb

Repository: github.com/once-ere/7plus1 — prepared for Nash
Notebook: 7+1.nb (Wolfram Language 15.0.1)
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Abstract

For each symbol this dictionary gives the mathematical notation, the name used in the Mathematica notebook 7+1.nb (where there is one; the names of section 25 begin with `tb`, those of section 26 with `rd`; "—" means that the notebook has no separate name), the defining equation and its meaning, grouped by topic in the order of the document. Longer defining equations follow each table as numbered displayed formulas. Section 14 lists the symbols that look alike, section 15 the Mathematica names. Every equation needed to define a symbol is written in this document.

Contents

1. Conventions **2**
2. The bulk, the torus and the brane **2**
3. Metrics and the action **3**
4. How the brane sits in the bulk; matter on the brane **4**
5. Gravity seen from the brane; the radion **5**
6. The Higgs boson and the radion **5**
7. The Standard Model on the brane **6**
8. dirac16complex at the brane **7**
9. The portals **7**
10. The three regimes **8**
11. The brane on a 4-sphere **8**
12. Warped branes (Randall–Sundrum) **9**
13. Constants and data **9**
14. Symbols that look alike **10**
15. Mathematica names **10**
15.1 Names of section 25 used by the document 10
15.2 Names of section 26 used by the document 11
15.3 Names of the earlier sections 11

1. Conventions

- Units: $\hbar = c = 1$; energies, masses and inverse lengths in GeV; $\hbar c = 1.973269804 \times 10^{-16}$ GeV m (Mathematica `tbHbarCm`).
- Coordinates $X = (x_0, x_1, \ldots, x_7)$ (Mathematica `X = {x0, …, x7}`): x_0 is time (the direction of the octonion unit $e_0 = 1$), x_1, x_2, x_3 ordinary space, $x_4, \ldots, x_7$ the four auxiliary space-like coordinates $\vec{y}$; on the brane $x = (x_0, \ldots, x_3)$.
- Indices: $A, B = 0, \ldots, 7$ (all directions); $\mu, \nu = 0, \ldots, 3$ (along the brane); $k, l = 4, \ldots, 7$ (transverse to it); $i, j = 1, 2, 3$ (ordinary space); $a = 1, \ldots, 4$ (the transverse spinor index). A prime is d/dx_0 (or d/dy for warped metrics), a dot d/ds.
- Curvature (mostly-minus signature): $R^{\rho}{}_{\sigma\mu\nu} = \partial_\mu \Gamma^{\rho}{}_{\nu\sigma} - \partial_\nu \Gamma^{\rho}{}_{\mu\sigma} + \Gamma^{\rho}{}_{\mu\lambda}\Gamma^{\lambda}{}_{\nu\sigma} - \Gamma^{\rho}{}_{\nu\lambda}\Gamma^{\lambda}{}_{\mu\sigma}$, $R_{\sigma\nu} = R^{\rho}{}_{\sigma\rho\nu}$, $R = g^{\sigma\nu} R_{\sigma\nu}$, $G^A{}_B = R^A{}_B - \frac{1}{2}\delta^A_B R$; $G^A{}_B - \Lambda \delta^A_B = \kappa T^A{}_B$.
- Status labels used in the document: an ESTIMATE is an order of magnitude with numerical factors omitted; everything else is an exact consequence of the definitions, checked by sections 25–26 of 7+1.nb and by tools/ref_torus_brane.py and tools/ref_radion.py.

The signature: $\{7, 1, 0\}$ and $\mathrm{diag}(+1, -1, \ldots, -1)$. The 7+1 spacetime has one time-like dimension (x_0, the direction of the octonion unit $e_0 = 1$) and seven space-like dimensions (x_1, x_2, x_3 ordinary space, $x_4, \ldots, x_7$ auxiliary space). Two different objects describe this, and neither may be replaced by the other. (i) The tensor package xAct, used in section 12 of 7+1.nb exactly as in the author's notebook Einstein-Lovelock-7+1-Nash.nb, declares the metric with `DefMetric[{7, 1, 0}, g71[-a, -b], CD, …]`: the list $\{7, 1, 0\}$ declares seven space-like dimensions, one time-like dimension and no degenerate dimension — the numbers of entries $+1$ and -1 of $\eta_{71} = \mathrm{diag}(-1, +1, \ldots, +1)$ (`η71`), the same spacetime written in the sign convention of 4+4.nb (space-like $= +1$, time-like $= -1$). (ii) The matrix $\eta = -\eta_{71} = \mathrm{diag}(+1, -1, \ldots, -1)$ (`η17`) is simply the squares of the octonion basis: $e_0^2 = +1$ for the unit $e_0 = 1$ (time) and $e_i^2 = -1$ for the seven imaginary units $e_1, \ldots, e_7$ (space); more generally it is the real part of the product of two basis units, and it gives the quadratic form of the octonions:

$$\eta_{AB} = \mathrm{Re}(e_A e_B), \qquad e_0^2 = +1, \qquad e_i^2 = -1 \quad (i = 1, \ldots, 7),$$
$$\mathrm{Re}(x\,x) = x_0^2 - x_1^2 - \cdots - x_7^2 = \eta_{AB}\, x^A x^B \qquad (x = x^A e_A \in \mathbb{O}).$$

η gives the signs of the Clifford relation $\Gamma_A \Gamma_B + \Gamma_B \Gamma_A = 2\,\eta_{AB}\, 1_{16}$ of the real 16×16 Dirac matrices: time squares to $+1_{16}$ and each of the seven space directions to -1_{16}, exactly like the octonion units (in the notation $Cl(p, q)$, with p generators squaring to $+1$ and q to -1, the algebra $Cl(1, 7) \cong \mathbb{R}(16)$). It also gives the signs of the metric components: $g_{00} > 0$ along the time-like direction, negative squares along the space-like directions. Every component computed by 7+1.nb follows from the component matrix of the metric; the declared list fixes for xAct the sign of the determinant of the metric, `SignDetOfMetric[g71]` $= -1$, which is the same in both sign conventions, since $\det \eta = \det \eta_{71} = -1$. 7+1.nb checks `SignatureOfMetric[g71]` $= \{7, 1, 0\}$, `SignDetOfMetric[g71]` $= -1$, $\mathrm{Re}(e_A e_B) = \eta_{AB}$ and $\mathrm{Re}(x\,x) = \eta_{AB} x^A x^B$.

2. The bulk, the torus and the brane

symbol	Mathematica	definition	meaning
$\vec{x}$, $\vec{y}$	`{x1, x2, x3}`, `tbY`	$\vec{y} = (x_4, \ldots, x_7)$	ordinary space; the auxiliary coordinates
L_c, n_k	`tbLc`	(2.1)	coordinate period; integers
T^4	—	(2.1)	the auxiliary torus
M	—	(2.1)	the bulk
b, b_0	`b4[x0]`, `tbB`	(2.1)	size of the torus; its value today
$V_\perp$, $\ell_\perp$	`tbV`, `rIIVp`, `tbLp`	(2.1)	proper volume and size of the torus
W	`tbOnW` (the rule $\vec{y} \to 0$)	(2.2)	the world brane
$X^A(x)$, $J^A{}_\mu$	`tbJ`	(2.2)	embedding of the brane; its Jacobian
x^μ	—	$x^\mu = X^\mu$	brane coordinates

symbol	Mathematica	definition	meaning
g_{AB}	`gCos`, `gPoly`	section 3 of this dictionary	bulk metric
$h_{\mu\nu}$	`tbInduced[g]`	(2.3)	induced metric of the brane
$e_0, \ldots, e_3$	—	(2.4)	basis of the quaternion subalgebra $\mathbb{H}$ of the octonions
ϵ_{ijk}	—	totally antisymmetric, $\epsilon_{123} = 1$	Levi-Civita symbol

$$x_k \simeq x_k + L_c n_k, \qquad T^4 = \mathbb{R}^4/(L_c\mathbb{Z})^4, \qquad M = \mathbb{R}_{x_0} \times \mathbb{R}^3_{\vec{x}} \times T^4_{\vec{y}}, \tag{2.1}$$

$$b^8 = \det(-g_{kl}), \qquad V_\perp = b^4 L_c^4, \qquad \ell_\perp = b L_c,$$

$$W = \{X \in M : \; x_4 = x_5 = x_6 = x_7 = 0\}, \tag{2.2}$$

$$X^A(x) = (x_0, x_1, x_2, x_3, 0, 0, 0, 0), \qquad J^A{}_\mu = \frac{\partial X^A}{\partial x^\mu},$$

$$h_{\mu\nu} = g_{AB} J^A{}_\mu J^B{}_\nu = g_{\mu\nu}(x, \vec{y} = 0), \qquad h = J^{\mathsf{T}} g J, \tag{2.3}$$

$$e_1 e_2 = -e_3, \quad e_2 e_3 = -e_1, \quad e_3 e_1 = -e_2, \quad e_i e_j = -\delta_{ij} - \epsilon_{ijk} e_k, \tag{2.4}$$

$$\mathrm{Re}(e_\mu e_\nu) = \mathrm{diag}(1, -1, -1, -1).$$

3. Metrics and the action

symbol	Mathematica	definition	meaning
n, a, b	`nP`, `aP`, `bP`; `a3[x0]`, `b4[x0]`	(3.1)	lapse, scale factors of ordinary space and of the torus
τ	—	time since the singularity	time of the primordial regime
p_1, q_1	`rIp1`, `rIq1`	(3.2)	Einstein Kasner exponents
S, S_{grav}, S_ψ, S_U, S_W	—	(3.3)	the action and its parts
κ, G_8, M_8	`tbKap`, `rIIM8`	$\kappa = 8\pi G_8 = M_8^{-6}$	eight-dimensional gravity
Λ	`lamb`	(3.3)	bulk cosmological constant
R, L_2, L_3, α_2, α_3	`alf[2]`, `alf[3]`	(3.4)	scalar curvature, Lovelock densities and couplings
ψ, $\bar{\psi}$	`dirac16complex`	$\bar{\psi} = \psi^\dagger \Gamma_0$	the 16-component bulk field and its conjugate
$\Gamma^A(X)$, Ω_A	—	frame Dirac matrices, spin connection	Dirac operator of the bulk
m	—	mass	mass of dirac16complex
S_{self}	—	nine contact terms	self-couplings of dirac16complex
$U(b)$, U_0, K_U	`uPot`, `u0B`, `kU`	(5.4)	stabilizing potential
f_W	`tbfW`, `fWb`	(4.4)	brane tension
$\mathcal{L}_{\mathrm{SM}}$, $\mathcal{L}_{\mathrm{portal}}$	—	sections 8, 9 of this dictionary	Standard Model; portals

$$ds^2 = n^2 dx_0^2 - a^2 d\vec{x}^2 - b^2 d\vec{y}^2 \qquad \text{(inhomogeneous example:)} \tag{3.1}$$

$$n = 1 + \tfrac{x_0 x_4}{5} + \tfrac{x_0^2}{11}, \qquad a = 1 + \tfrac{x_0}{3} + \tfrac{x_4^2}{7}, \qquad b = 2 - \tfrac{x_0 x_4}{4} + \tfrac{x_0^2}{9} + \tfrac{x_4}{13},$$

$$p_1 = \frac{1 + 2\sqrt{2}}{7}, \qquad q_1 = \frac{2 - 3\sqrt{2}}{14}, \qquad a \propto \tau^{p_1}, \; b \propto \tau^{q_1}, \tag{3.2}$$

$$S = S_{\mathrm{grav}} + S_\psi + S_U + S_W, \qquad S_{\mathrm{grav}} = -\frac{1}{2\kappa} \int d^8X \sqrt{|g|}\, (2\Lambda + R + \alpha_2 L_2 + \alpha_3 L_3), \tag{3.3}$$

$$S_W = \int_W d^4x \sqrt{|h|}\, (-f_W + \mathcal{L}_{\mathrm{SM}} + \mathcal{L}_{\mathrm{portal}}),$$

$$L_2 = R^2 - 4R_{AB}R^{AB} + R_{ABCD}R^{ABCD}, \tag{3.4}$$

$$L_3 = R^3 - 12RR_{AB}R^{AB} + 16R^A{}_B R^B{}_C R^C{}_A + 24R^{AC}R^{BD}R_{ABCD} + 3RR_{ABCD}R^{ABCD}$$
$$- 24R^{AB}R_A{}^{CDE}R_{BCDE} + 4R^{AB}{}_{CD}R^{CD}{}_{EF}R^{EF}{}_{AB} - 8R_A{}^C{}_B{}^D R_C{}^E{}_D{}^F R_E{}^A{}_F{}^B.$$

4. How the brane sits in the bulk; matter on the brane

symbol	Mathematica	definition	meaning
$n_{(k)}$	—	$b^{-1}\partial_k$	unit normals of the brane
$K^{(k)}_{\mu\nu}$	—	(4.1)	extrinsic curvature
$\{\lambda\,\mu\nu\}$	`tbChrW`	(4.1)	Christoffel symbols
s	—	proper time	parameter of a geodesic
$G_\perp$	—	(4.2)	Green's function of the four transverse directions
$\delta^4(\vec{y})$, F	`DiracDelta`	(4.3)	delta function of the torus; a test function
$\vec{q}_{\vec{n}}$	`tbQ[n]`	$2\pi\vec{n}/L_c$	wave vector of a mode
$T^W_{\mu\nu}$, T^{AB}	—	(4.4)	brane energy–momentum; its eight-dimensional source
ε, p, w	—	(4.4)	energy density, pressure, $w = p/\varepsilon$
p_a, p_b	`feStress[W]`	(4.5)	pressures along ordinary space and along the torus
$W(a,b)$	argument of `feStress`	(4.5)	comoving energy of a unit coordinate cell
H_a, H_b	`Ha`, `Hb`	a'/a, b'/b	expansion rates
ρ_4, p_4	`tbRho`	(4.5)	energy density and pressure of a brane fluid
k_a, k_b, ω, θ	`tbAa`, `tbBb`, `tbth`	(4.6)	comoving momenta, energy and angle of a massless particle
$u_0, \ldots, u_3$	`tbu0` … `tbu3`	$h = \mathrm{diag}(1/u_0, -1/u_1, -1/u_2, -1/u_3)$	components used in the tension check

$$K^{(k)}_{\mu\nu} = \frac{1}{2}(\mathcal{L}_{n_{(k)}} g)_{\mu\nu} = \frac{1}{2b}\partial_k g_{\mu\nu}\Big|_W, \tag{4.1}$$

$$\{\lambda\,\mu\nu\} = \frac{1}{2} g^{\lambda\sigma}(\partial_\mu g_{\sigma\nu} + \partial_\nu g_{\sigma\mu} - \partial_\sigma g_{\mu\nu}),$$

$$\nabla^2_{\vec{y}} \frac{1}{|\vec{y}|^2} = -4\pi^2\delta^4(\vec{y}), \qquad G_\perp = -\frac{1}{4\pi^2|\vec{y}|^2}, \tag{4.2}$$

$$\int_W d^4x\sqrt{|h|}\,\mathcal{L} = \int d^8X\sqrt{|g|}\,\frac{\delta^4(\vec{y})}{b^4}\,\mathcal{L}, \qquad \delta^4(\vec{y}) = \frac{1}{L_c^4}\sum_{\vec{n}} e^{i\vec{q}_{\vec{n}}\cdot\vec{y}}, \tag{4.3}$$

$$T^W_{\mu\nu} = \frac{2}{\sqrt{|h|}}\frac{\delta S_W}{\delta h^{\mu\nu}}, \qquad T^{AB} = \frac{\delta^4(\vec{y})}{b^4} T^{\mu\nu}_W \delta^A_\mu \delta^B_\nu, \tag{4.4}$$

$$\mathcal{L}_W = -f_W \;\Rightarrow\; T^W_{\mu\nu} = f_W h_{\mu\nu}, \quad \varepsilon = f_W, \quad p = -f_W, \quad w = -1,$$
$$\varepsilon = \frac{W}{a^3b^4}, \qquad p_a = -\frac{\partial_a W}{3a^2b^4}, \qquad p_b = -\frac{\partial_b W}{4a^3b^3}, \tag{4.5}$$
$$\text{tension: } W = \frac{a^3 f_W}{L_c^4}; \qquad \text{brane fluid: } W = \frac{a^3\rho_4}{L_c^4},$$

$$W = \omega = \sqrt{k_a^2/a^2 + k_b^2/b^2}, \qquad k_a/a = p\cos\theta, \quad k_b/b = p\sin\theta, \tag{4.6}$$
$$\text{isotropic in seven dimensions: } p_a = p_b = \frac{\varepsilon}{7}.$$

5. Gravity seen from the brane; the radion

symbol	Mathematica	definition	meaning
G_4, $M_{\rm Pl}$	`tbGN`, `tbMP`, `rIIMPl`	(5.1)	four-dimensional Newton constant; reduced Planck mass
Λ_4	—	(5.1)	four-dimensional cosmological constant
H_0, Ω_m, Ω_r, Ω_Λ, t_0	`rIIH0g`, `rIIOL`	(5.2)	Hubble rate, density fractions, age today
ρ_Λ	`dkRhoL`	$3H_0^2 M_{\rm Pl}^2 \Omega_\Lambda$	observed dark-energy density
Φ_8, $\Phi_{\rm N}$	—	(5.3)	Newtonian potentials (eight-dimensional; on the torus)
M, M', r	`tbr`	—	two masses on the brane; their distance
β, ϕ	`rdBt`, `rdBe`	(5.4)	the radion; the canonical radion
g_E	—	(5.4)	four-dimensional Einstein-frame metric
m_r	—	10 GeV	radion mass
$T^\mu{}_\mu$, $\mathcal{L}_{\phi W}$, V_ϕ	—	(5.4)	trace of the brane energy–momentum; radion coupling; radion potential
$\Gamma_{\rm brane}$, $\Gamma_{\rm bulk}$	—	section 22.2 of 7+1.nb	radion decay rates into brane fields and into the bulk
Q, Q'	—	$V = QQ'/(4\pi r)$	electric charges on the brane

$$3H_a^2 = 8\pi G_4 \rho_4 + \Lambda_4, \qquad H_a' = -\frac{\kappa}{2}(\varepsilon + p_a), \tag{5.1}$$

$$8\pi G_4 = \frac{\kappa}{V_\perp} = \frac{1}{M_{\rm Pl}^2}, \qquad \Lambda_4 = \Lambda + \kappa U_0 + \frac{\kappa f_W}{V_\perp},$$

$$t_0 = \frac{1}{H_0}\int_0^1 \frac{da}{a\sqrt{\Omega_r a^{-4} + \Omega_m a^{-3} + \Omega_\Lambda}}, \tag{5.2}$$

$$\Phi_8 = -\frac{5\kappa M}{32\pi^3 r^5}, \qquad \Phi_{\rm N} = -\frac{5\kappa M}{32\pi^3}\sum_{\vec n}\frac{1}{(r^2 + \ell_\perp^2 |\vec n|^2)^{5/2}}, \tag{5.3}$$

$$\Phi_{\rm N} = -\frac{5}{3}\frac{G_4 M}{r}\sum_{\vec m} e^{-2\pi|\vec m| r/\ell_\perp},$$

$$\beta = \ln\frac{b}{b_0}, \qquad U(b) = U_0 + \frac{K_U}{2}\ln^2\frac{b}{b_0}, \qquad m_r^2 = \frac{\kappa K_U}{12}, \qquad \phi = \sqrt{12} M_{\rm Pl}\beta, \tag{5.4}$$

$$h = e^{-4\beta} g_E, \qquad \mathcal{L}_{\phi W} = \frac{\phi\, T^\mu{}_\mu}{\sqrt{3} M_{\rm Pl}}, \qquad V_\phi = -\frac{2}{3}\frac{G_4 M M'}{r} e^{-m_r r}.$$

6. The Higgs boson and the radion

symbol	Mathematica	definition	meaning
Φ, $\tilde\Phi$	—	$\tilde\Phi = i\sigma_2\Phi^*$	Higgs doublet; its conjugate
φ_H, h	`rdPhE`, `rdHh`	$\Phi^\dagger\Phi = \varphi_H^2/2$, $\varphi_H = v + h$	radial Higgs field; the Higgs field h (not the metric $h_{\mu\nu}$)
v	`vEW`	$(\sqrt{2}G_F)^{-1/2}$	Higgs vacuum value
λ	`lamH`, `rdLamH`	$m_H^2/(2v^2)$	Higgs quartic coupling
ξ	`rdXi`	(6.1)	curvature coupling of the Higgs ($\frac{1}{6}$: conformal)
$R(h)$	`rdRh`	scalar curvature of $h_{\mu\nu}$	curvature of the brane metric
Ω	`rdOmg`, `rdOmE`	(6.2)	conformal factor of the brane metric
$\Box$	—	$\partial_0^2 - \nabla^2$	flat wave operator
γ	`rdGam`	$v/(\sqrt{3}M_{\rm Pl})$	Higgs–radion mixing parameter

symbol	Mathematica	definition	meaning
m_{h0}, m_{r0}	`rdMh0`, `rdMr0`	$m_{h0}^2 = 2\lambda v^2$	bare Higgs and radion masses
h', ϕ', Z	`rdSm`, `rdZ`	(6.3)	shifted fields; normalization
θ, h_m, ϕ_m	`rdth`, `rdQ`	(6.3)	mixing angle; mass eigenstates

$$\mathcal{L} = \sqrt{|h|}\Big[(D_\mu\Phi)^\dagger D^\mu\Phi - \lambda\Big(\Phi^\dagger\Phi - \frac{v^2}{2}\Big)^2 + \xi R(h)\Phi^\dagger\Phi\Big] + \frac{1}{2}(\partial\phi)^2 - \frac{1}{2}m_{r0}^2\phi^2, \tag{6.1}$$

$$h_{\mu\nu} = \Omega^2 g_{E\mu\nu}, \qquad \Omega = e^{-2\beta} = e^{-\phi/(\sqrt{3}M_{\rm Pl})}, \qquad \sqrt{|h|}R(h) = -6\Omega\Box\Omega \;\; (g_E = \eta), \tag{6.2}$$

$$h = h' - \frac{6\xi\gamma}{Z}\phi', \qquad \phi = \frac{\phi'}{Z}, \qquad Z^2 = 1 + 6\xi\gamma^2(1-6\xi), \tag{6.3}$$

$$h' = \cos\theta\, h_m - \sin\theta\,\phi_m, \qquad \phi' = \sin\theta\, h_m + \cos\theta\,\phi_m, \qquad \tan 2\theta = \frac{12\gamma\xi Z m_{h0}^2}{m_{r0}^2 - m_{h0}^2(Z^2 - 36\xi^2\gamma^2)}.$$

7. The Standard Model on the brane

symbol	Mathematica	definition	meaning
G_μ^a, W_μ^I, B_μ	—	gauge fields	$SU(3)$ ($a = 1..8$), $SU(2)$ ($I = 1,2,3$), $U(1)_Y$
g_s, g, g'	`gEW`, `gpEW`	$g = 2m_W/v$, $g' = g\tan\theta_W$	gauge couplings
Q, T_3, Y	—	$Q = T_3 + Y$	electric charge, weak isospin, hypercharge
Q_L, u_R, d_R, L_L, e_R, ν_R	—	table of section 8 of the document	the Weyl fields of one generation
g_*, g_{*s}	—	$28 + \frac{7}{8}\cdot 90 = 106.75$	relativistic degrees of freedom (energy, entropy)
$\gamma^\mu(x)$	—	$\{\gamma^\mu, \gamma^\nu\} = 2h^{\mu\nu}$	Dirac matrices of the brane metric
D_μ, ∇_μ	—	(7.1)	gauge-covariant derivative; covariant derivative of h
T^a, τ^I, f^{abc}	—	(7.1)	$SU(3)$ generators, $\tau^I = \sigma_I$, structure constants
Y_u, Y_d, Y_e, Y_ν	—	3×3 matrices	Yukawa matrices
μ^2	—	$-\lambda v^2$	Higgs mass parameter
G_F, m_W, m_Z, m_H	`gFermi`, `mWb`, `mZb`, `mHb`	PDG 2025	inputs
θ_W, e	`sW2`, `eEW`	$\sin^2\theta_W = 1 - m_W^2/m_Z^2$, $e = g\sin\theta_W$	weak mixing angle; electric charge

$$D_\mu = \nabla_\mu - ig_s G_\mu^a T^a - igW_\mu^I\frac{\tau^I}{2} - ig'YB_\mu, \qquad G_{\mu\nu}^a = \partial_\mu G_\nu^a - \partial_\nu G_\mu^a + g_s f^{abc}G_\mu^b G_\nu^c, \tag{7.1}$$

$$\mathcal{L}_{\rm SM} = -\frac{1}{4}G_{\mu\nu}^a G^{a\mu\nu} - \frac{1}{4}W_{\mu\nu}^I W^{I\mu\nu} - \frac{1}{4}B_{\mu\nu}B^{\mu\nu} + (D_\mu\Phi)^\dagger D^\mu\Phi - \mu^2\Phi^\dagger\Phi - \lambda(\Phi^\dagger\Phi)^2 + \sum_f \bar f i\gamma^\mu D_\mu f - [\text{Yukawa terms} + \text{h.c.}].$$

8. dirac16complex at the brane

symbol	Mathematica	definition	meaning
Γ_A, Γ^A	`Γ17[A]`, `Γ17up[A]`	(8.1)	real 16×16 Dirac matrices of 7+1
σ_i	`PauliMatrix[i]`	(8.1)	Pauli matrices
γ_μ, γ_5	`g4D`, `g5D`	(8.1)	4×4 Dirac matrices of the brane
ρ_k	`rhoK`	(8.1)	anti-Hermitian 4×4 matrices of the transverse directions
R_A	`rBr`	(8.1)	brane-adapted 16×16 representation
U	`uInt`, `tbU8` ($2\sqrt{2}U$)	(8.2)	unitary intertwiner
$\Gamma^{(K)}$, $R^{(K)}$, s	—	(8.2)	Clifford products; a matrix unit
ζ, ψ_a, e_a	—	(8.2)	brane-adapted field; the four brane spinors; unit vectors of $\mathbb{C}^4$
u	`tbu`	$(1,2,0,-1)$	the brane spinor of Example 9.2
ρ_{4567}, P_s	`rho4567`, `tbPs`	(8.3)	transverse chirality; its projectors
χ, P_L	`PL17`	(8.3)	eight-dimensional chirality; left projector
X (operator), $m_{\vec n}$, $\chi_{\vec n}$	`xKK`	(8.4)	Kaluza–Klein operator; masses; four-dimensional fields
$r_4(k)$, $\sigma(k)$	`rIIr4`, `rIIsig`	$8\sigma(k)-32\sigma(k/4)$	lattice multiplicity; divisor sum

$$\Gamma_A\Gamma_B+\Gamma_B\Gamma_A=2\eta_{AB}1_{16},\qquad \gamma_0=\mathrm{diag}(1,1,-1,-1), \tag{8.1}$$

$$\gamma_i=\begin{pmatrix}0&\sigma_i\\-\sigma_i&0\end{pmatrix},\qquad \gamma_5=i\gamma_0\gamma_1\gamma_2\gamma_3,$$

$$\rho_4=i\gamma_0,\quad \rho_{5,6,7}=-\gamma_{1,2,3},\qquad R_\mu=\gamma_\mu\otimes1_4,\quad R_k=\gamma_5\otimes\rho_k,$$

$$\Gamma_AU=UR_A,\qquad U\propto\sum_K\Gamma^{(K)}s\,(R^{(K)})^{-1},\qquad \psi=U\zeta,\qquad \zeta=\sum_{a=1}^{4}\psi_a\otimes e_a, \tag{8.2}$$

$$\rho_{4567}=\rho_4\rho_5\rho_6\rho_7=-\gamma_5,\qquad P_s=\tfrac12(1_4+s\rho_{4567}), \tag{8.3}$$

$$\chi=i\Gamma_0\cdots\Gamma_7=U(\gamma_5\otimes\rho_{4567})U^\dagger,\qquad P_L=\tfrac12(1-\chi),$$

$$X=\gamma_5\otimes\frac{\vec\rho\cdot\vec q}{b_0},\qquad X^2=-\frac{|\vec q|^2}{b_0^2},\qquad m_{\vec n}=\sqrt{m^2+\frac{4\pi^2|\vec n|^2}{\ell_\perp^2}}, \tag{8.4}$$

$$\psi(x,0)=\frac{1}{\sqrt{V_\perp}}\sum_{\vec n}\chi_{\vec n}(x).$$

9. The portals

symbol	Mathematica	definition	meaning
y^L_{is}, y^R_{is}, y	`cyS` (c_y)	(9.1)	Yukawa (neutrino) portal couplings
$\psi^{(L)}_s$, $\psi^{(R)}_s$, P_R	—	$P_R=\frac12(1+\gamma_5)$	brane-right-handed parts of the chiral halves
μ_{is}, μ_P	`dkMuP`	(9.1)	portal Dirac masses
G^f_P, G_P	`cVS` (c_V)	(9.1)	vector portal
λ_P	`cPS` (c_P)	(9.1)	Higgs portal
$\phi_{\vec n}$	`tbPhi[n]`	$e^{i\vec q_{\vec n}\cdot\vec y}/\sqrt{V_\perp}$	normalized modes of the torus
c_P, c_y, c_V	`cPS`, `cyS`, `cVS`	(9.2)	couplings in units of M_8
$\lambda_{\rm eff}$, $y_{\rm eff}$, $G_{\rm eff}$	`dkLam`, `dkYeff`	(9.2)	four-dimensional couplings of one level
$\mathcal{M}$, m_D, N	—	(9.3)	neutrino mass matrix; Dirac masses; light mixing block

symbol	Mathematica	definition	meaning
$U_{\rm PMNS}$, m_1, m_2, m_3	—	section 17 of 7+1.nb	lepton mixing matrix; light neutrino masses

$$\mathcal{L}_{\rm portal} \supset -\sum_{i,s}\left[y^L_{is}\bar{L}_i\tilde{\Phi}P_R\psi^{(L)}_s + y^R_{is}\bar{L}_i\tilde{\Phi}P_R\psi^{(R)}_s\right] + \text{h.c.} \tag{9.1}$$
$$- G^f_P(\bar{\psi}\Gamma^\mu\psi)(\bar{f}\gamma_\mu f) - \lambda_P\Phi^\dagger\Phi\,\bar{\psi}\psi, \qquad \mu_{is} = \frac{y_{is}v}{\sqrt{2}\sqrt{V_\perp}},$$

$$\lambda_P = \frac{c_P}{M_8^5}, \qquad y = \frac{c_y}{M_8^2}, \qquad G_P = \frac{c_V}{M_8^6}, \tag{9.2}$$
$$\lambda_{\rm eff} = \frac{c_P M_8}{M_{\rm Pl}^2}, \qquad y_{\rm eff} = \frac{c_y M_8}{M_{\rm Pl}}, \qquad G_{\rm eff} = \frac{c_V}{M_{\rm Pl}^2}, \qquad \mu_P = \frac{y_{\rm eff}v}{\sqrt{2}},$$

$$\mathcal{M} = \begin{pmatrix} m_D & \mu \\ 0 & m1_4 \end{pmatrix}, \qquad m_D = U_{\rm PMNS}\mathrm{diag}(m_1,m_2,m_3), \qquad 1 - NN^\dagger \simeq \frac{\mu\mu^\dagger}{m^2}. \tag{9.3}$$

10. The three regimes

symbol	Mathematica	definition	meaning
T	—	temperature	temperature of the brane plasma
T_*	`tbTstar`	$(M_8^6/M_{\rm Pl})^{1/5}$	normalcy temperature (ESTIMATE)
s (entropy)	—	$\frac{2\pi^2}{45}g_{*s}T^3$	entropy density
t, z	—	cosmic time; redshift	—
H_∞	`rIIIH`	$H_0\Omega_\Lambda^{1/2}$	final de Sitter rate

11. The brane on a 4-sphere

symbol	Mathematica	definition	meaning
S^4, r, r_0	`rdR0`	(11.1)	the 4-sphere; its radius; the reference radius
$\hat{\gamma}$	`rdGam4`	(11.1)	metric of the unit 4-sphere
$X_1, \ldots, X_5$	`rdEmb`	(11.1)	embedding in $\mathbb{R}^5$; the north pole $(0,0,0,0,r)$
L_{ab}	—	$X_a\partial_b - X_b\partial_a$	the 10 rotations ($SO(5)$)
l, μ_l, d_l	`rdDl`	(11.2)	degree, mass and multiplicity of a level
Z_l, $C_l^{(3/2)}$	`rdZon`	(11.2)	zonal harmonic; Gegenbauer polynomial
r_{S^4}	`rdRS4`	$(3V_\perp/(8\pi^2))^{1/4}$	radius of a sphere of volume $V_\perp$
$\tilde{\gamma}_a$, e^a, ε_s	`rdGt`, `rdHf`, `rdKS`	(11.3)	Dirac matrices, frame and Killing spinors of S^4
λ (Dirac), $R_{\rm int}$	—	(11.3)	Dirac eigenvalues; $R_{\rm int} = 12/r^2$
U_1	`rdU1`	$U = U_0 + U_1\beta + \frac{K_U}{2}\beta^2$	slope of the stabilizing energy (sphere)

$$g_{kl} = -r^2\hat{\gamma}_{kl}, \qquad \hat{\gamma} = \mathrm{diag}(1, \sin^2 x_4, \sin^2 x_4 \sin^2 x_5, \sin^2 x_4 \sin^2 x_5 \sin^2 x_6), \qquad V = \frac{8\pi^2}{3}r^4, \tag{11.1}$$
$$X_5 = r\cos x_4, \qquad X_4 = r\sin x_4 \cos x_5, \qquad X_3 = r\sin x_4 \sin x_5 \cos x_6,$$

$$(X_1, X_2) = r \sin x_4 \sin x_5 \sin x_6 \, (\cos x_7, \sin x_7),$$

$$\mu_l = \frac{\sqrt{l(l+3)}}{r}, \qquad d_l = \frac{(2l+3)(l+1)(l+2)}{6}, \tag{11.2}$$

$$Z_l = \sqrt{\frac{d_l}{V}} \frac{C_l^{(3/2)}(\cos x_4)}{C_l^{(3/2)}(1)}, \qquad C_l^{(3/2)}(1) = \frac{(l+1)(l+2)}{2},$$

$$\tilde{\gamma}_a = i\rho_{a+3}, \qquad e^a = (r, r\sin x_4, \dots) dx^{a+3}, \qquad \nabla_i \varepsilon_s = s \frac{i}{2r} e_i{}^a \tilde{\gamma}_a \varepsilon_s, \tag{11.3}$$

$$\lambda = \pm \frac{i(l+2)}{r}, \qquad |\lambda|^2 \geq \frac{n R_{\text{int}}}{4(n-1)} = \frac{4}{r^2}.$$

12. Warped branes (Randall–Sundrum)

symbol	Mathematica	definition	meaning
$A(y)$, k, y, R	`rdA`, `rdk`, `rdy`, `rdRR`	(12.1)	warp function, AdS scale, fifth coordinate, radius
f_{UV}, f_{IR}	`rdfUV`, `rdfIR`	(12.1)	brane tensions
$K^\mu{}_\nu$, $S_{\mu\nu}$	`rdKmix`, `rdKjump`	(12.1)	extrinsic curvature; brane energy–momentum
M (RS)	—	$\kappa = 1/(4M^3)$	five-dimensional scale of Randall and Sundrum
$T(x)$, μ, f, $\langle\mu\rangle$, $\delta\mu$	`rdT`, `rdmu`, `rdfm`, `rdmu0`	(12.2)	radius field; canonical radion; its vacuum value and fluctuation
v_A, v_B, m, ϵ	`rdvA`, `rdvB`, `rdMg`, `rdeps`	(12.2)	Goldberger–Wise brane values, mass, small parameter
$\vec{z}$	`rdz1` … `rdz3`	—	the three unwarped extra directions of the eight-dimensional test

$$ds^2 = e^{-2A(y)} \eta - dy^2, \qquad G^\mu{}_\nu = 3A'' - 6A'^2, \qquad G^y{}_y = -6A'^2, \tag{12.1}$$

$$f_{\text{UV}} = -f_{\text{IR}} = \frac{6k}{\kappa}, \qquad K^\mu{}_\nu = \frac{1}{2} g^{\mu\alpha} \partial_y g_{\alpha\nu} = -A' \delta^\mu_\nu,$$

$$\mu = f e^{-k\pi T}, \qquad f^2 = \frac{6}{\kappa k}, \qquad M_{\text{Pl}}^2 = \frac{1 - e^{-2\pi k R}}{\kappa k}, \qquad \mu = \langle\mu\rangle + \delta\mu, \tag{12.2}$$

$$m_r^2 = \frac{4}{3} \kappa k^2 \epsilon^2 v_B^2 \Big(\frac{\langle\mu\rangle}{f} \Big)^2, \qquad \epsilon = \frac{m^2}{4k^2}.$$

13. Constants and data

symbol	Mathematica	value	meaning and source
M_8	`rIIM8`	10^4 GeV	choice of sections 22–24
M_{Pl}	`rIIMPl`	2.435×10^{18} GeV	reduced Planck mass
m_r	—	10 GeV	radion mass (choice of sections 22–24)
m_h, m_H	`mHb`	125.20 GeV	Higgs mass (PDG 2025)
$\hbar c$	`tbHbarCm`	$1.973269804 \times 10^{-16}$ GeV m	CODATA 2018
H_0, Ω_m, Ω_r	—	67.4 km/s/Mpc, 0.315, 9.14×10^{-5}	Planck 2018
$\langle\mu\rangle$	—	3 TeV	the warped example of the author's note

14. Symbols that look alike

- h: the induced metric $h_{\mu\nu}$ (always with indices or as a matrix), the Higgs field h of section 6 (and h', h_m).
- Φ: the Higgs doublet Φ; Φ_8, Φ_{N} are gravitational potentials; φ_H the radial Higgs field.
- ϕ: the canonical radion ϕ (and ϕ', ϕ_m) and the Fourier modes $\phi_{\vec{n}}$ (always with an index).
- χ: the eight-dimensional chirality χ and the four-dimensional fields $\chi_{\vec{n}}$ (with an index).
- λ: the Higgs quartic λ, the Higgs portal λ_P, λ_{eff}, the Dirac eigenvalue λ on S^4, and the index of the Christoffel symbol $\{\lambda\,\mu\nu\}$.
- μ: the index $\mu = 0, \ldots, 3$; the portal masses μ_{is}, μ_P, μ; the Kaluza–Klein masses μ_l; the warped radion μ; μ^2 of the Higgs potential.
- γ: the brane Dirac matrices γ_μ, $\gamma^\mu(x)$, γ_5; the mixing parameter γ; $\hat{\gamma}$ and $\tilde{\gamma}_a$ of the sphere.
- ρ: the densities ρ_4, ρ_Λ; the matrices ρ_k, ρ_{4567}.
- σ: the Pauli matrices σ_i and the divisor sum $\sigma(k)$.
- r: the distance r between two masses; the radius r of the sphere (r_0, r_{S^4}); $r_4(k)$.
- R: the scalar curvature R, $R(h)$, R_{int}; the Randall–Sundrum radius R; the matrices R_A, $R^{(K)}$.
- K: the extrinsic curvatures $K^{(k)}_{\mu\nu}$, $K^\mu{}_\nu$; the stiffness K_U; the index K of a Clifford product.
- S: the action and its parts; the brane energy–momentum $S_{\mu\nu}$ of section 12; the sphere S^4; s is a sign or the proper time or the entropy.
- M: the bulk M; masses M, M'; the scales M_8, M_{Pl}, M (RS); the mass matrix $\mathcal{M}$.
- U: the intertwiner U; the stabilizing potential $U(b)$ with U_0, U_1; U_{PMNS}.
- W: the brane W; the comoving energy $W(a,b)$; the gauge fields W^I_μ.
- Q: electric charges Q, Q' and the quark doublet Q_L.
- f: the brane tension f_W and f_{UV}, f_{IR}; the normalization f of the warped radion; Standard Model fermions f; f^{abc}.
- ϵ, ε: the Levi-Civita symbol ϵ_{ijk} and the Goldberger–Wise parameter ϵ; the energy density ε and the Killing spinors ε_s.

15. Mathematica names

15.1 Names of section 25 used by the document

name	meaning
`tbY`, `tbLc`, `tbB`, `tbV`, `tbLp`, `tbHbarCm`	$\vec{y}$; L_c; b; $V_\perp$; $\ell_\perp$; $\hbar c$
`tbJ`, `tbOnW`, `tbInduced`, `tbPt`, `tbChrW`	the Jacobian J; the rule $\vec{y} \to 0$; $h = J^{\mathsf{T}} g J$; the point $x_0 = \frac{1}{2}$, $x_4 = 0$; Christoffel symbols at the brane
`tbQ`, `tbPhi`, `tbVtx`, `tbRho`	$\vec{q}_{\vec{n}}$; $\phi_{\vec{n}}$; overlap of three modes; ρ_4
`tbKap`, `tbGN`, `tbr`, `tbMP`, `tbS`, `tbSkk`	symbols κ, G, r, M_{Pl}; the image and Kaluza–Klein sums
`tbSPermQ`, `tbU8`, `tbUd`, `tbKp`, `tbZ16`	signed-permutation test; $2\sqrt{2}U$; $U^\dagger$; `KroneckerProduct`; the 16×16 zero matrix
`tbPs`, `tbEven`	P_s; the even permutations of $(4,5,6,7)$
`tbu`, `tbPsi`, `tbBil`	$u = (1,2,0,-1)$; $\psi = U(u \otimes e_1)$; $\bar{\psi}\Gamma^\mu\psi$
`tbfW`, `tbu0` … `tbu3`, `tbAa`, `tbBb`, `tbr0`, `tbth`	f_W; components of $h^{\mu\nu}$; arguments a, b of $W(a,b)$; a radiation constant; θ
`tbT`, `tbM8`, `tbTstar`, `tby`	symbols T, M_8; T_*; $\lvert\vec{y}\rvert$

15.2 Names of section 26 used by the document

name	meaning
rdBt, rdBe, rdR0, rdGam4, rdEmb, rdDl, rdZon, rdRS4	$\beta(x_0)$, β, r_0, $\hat{\gamma}$, X_a, d_l, $C_l^{(3/2)}(\cos x_4)$, r_{S^4}
rdGt, rdHf, rdKS, rdU1	$\tilde{\gamma}_a$; frame lengths; Killing spinors; U_1
rdA, rdk, rdy, rdRR, rdfUV, rdfIR, rdKmix, rdKjump	RS1: $A(y)$, k, y, R, tensions, $K^\mu{}_\nu$ and its jump
rdT, rdmu, rdfm, rdmu0, rdvA, rdvB, rdMg, rdeps, rdz1 … rdz3	the warped radion and its Goldberger–Wise stabilization; the unwarped directions
rdPhE, rdHh, rdLamH, rdXi, rdRh, rdOmg, rdOmE, rdGam	φ_H, h, λ, ξ, $R(h)$, Ω, γ
rdMh0, rdMr0, rdSm, rdZ, rdth, rdQ	m_{h0}, m_{r0}, the shift, Z, θ, the rotation

15.3 Names of the earlier sections

name	section	meaning
X, ID4, ID16	1	coordinates; unit matrices
Γ17[A], Γ17up[A], η17	3–5	Dirac matrices of 7+1, raised; η
gCos, a3[x0], b4[x0], cvCos, gPoly, nP, aP, bP, toH, Ha, Hb	11	the cosmological and inhomogeneous metrics, curvature, expansion rates
g4D, g5D, rhoK, rBr, uInt, rho4567, PL17	17.2	γ_μ, γ_5, ρ_k, R_A, U, ρ_{4567}, P_L
vEW, gFermi, mWb, mZb, mHb, gEW, gpEW, sW2, eEW, lamH	17.3	electroweak constants
xKK	18.1	the Kaluza–Klein operator X
feStress, uPot, u0B, kU, lamb, fWb, alf	20	stress of a comoving energy; stabilizing potential; Λ; f_W; Lovelock couplings
rIp1, rIq1	21.1	Kasner exponents
rIIr4, rIIsig, rIIVp, rIIM8, rIIMPl, rIIH0g, rIIOL	22	lattice sums; $V_\perp$, M_8, $M_{\rm Pl}$; H_0, Ω_Λ
rIIIH	23	H_∞
cPS, cyS, cVS, dkLam, dkYeff, dkMuP, dkRhoL	24	portal couplings; $\lambda_{\rm eff}$, $y_{\rm eff}$, μ_P; ρ_Λ